

OpenAttack: An Open-Source Framework for Textual Adversarial Attack

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Abstract—Adversarial attacks on NLP models have shown that there are important weaknesses in the deep learning models used today. Therefore, this paper will evaluate the effectiveness of the OpenAttack package by testing its efficiency on several datasets, victim models, and different types of attacks. In doing so, we will measure the effectiveness of different attacks based on their influence on the accuracy of the models under test in terms of attack success rate, semantic similarity, and grammatical correctness.

Aside from testing some of the attacks available in OpenAttack, we will explore some robustness studies on defense mechanisms such as adversarial training and input pre-processing techniques to defend against such adversarial attacks. On top of that, this paper proposes a sentence-level adversarial attack which creates semantically consistent but misleading sentences that reduce the accuracy of the models under consideration while retaining high-quality linguistic output.

Our attack will be tested on several sentence-level attacks available in OpenAttack, comparing our results with those obtained using other attacks.

I. INTRODUCTION

It is fair to say that deep neural networks have completely transformed NLP tasks, obtaining the best results for various tasks including sentiment analysis, machine translation, question answering, and text classification. Deep transformer models are known for the ability to grasp complex context-based interactions and semantic representations in languages. However, recently there have been multiple studies uncovering serious weaknesses in such approaches, namely their vulnerability to adversarial attacks—the specially designed perturbations applied to the input text, making it unrecognizable by the model while preserving the meaning of the text [2].

In a formal setting, if the input sequence $\mathbf{x}$ is classified by a victim model F into a correct class y , an adversarial example is defined as a perturbation $\mathbf{x}'$ on $\mathbf{x}$ such that:

$$y_{\text{true}} = y \quad \text{but} \quad F(\mathbf{x}') = y' \neq y \quad (1)$$

Whereas perturbations in continuous spaces, like that of computer vision, face fewer restrictions, those in natural language processing (NLP) have to comply with multiple conditions. First, they should retain linguistic coherence. They should also be semantically similar to the original input, thus preserving the same meaning. Moreover, they should not be perceptible to humans, which is a challenge in itself. As such, crafting adversarial attacks on NLP models is more

complicated and necessitates the use of techniques such as synonym substitution and paraphrasing.

The potential impact of such an attack is profound. In practice, these samples could be used to bypass content moderation algorithms, distort sentiment classification models, circumvent spam and phishing filters, and influence decision-making processes. In particular, adversarial samples could be generated from malicious content to avoid detection. Therefore, as NLP models become more widely deployed across different sectors, it is crucial to ensure that they can withstand adversarial perturbations.

In order to tackle these issues, a number of adversarial attack techniques have been developed at various granularities, such as character-level attacks, which make small typographic changes, word-level attacks, where words are replaced by synonyms, and sentence-level attacks, where more advanced operations like paraphrasing are carried out. As a response, various types of defenses have also been considered, from adversarial training to data preprocessing and regularization. Many of these methods are restricted in their applicability and do not work with a variety of attacks.

OpenAttack has become an open-source library for performing adversarial research in NLP tasks. The library is able to provide support for the implementation, comparison, and evaluation of many attack methods and defense strategies on common metrics. OpenAttack is capable of using attacks with various granularities, can be combined with different machine learning libraries, and uses standard metrics for evaluating its performance. Despite all its capabilities, systematic studies demonstrating the effectiveness of the library are currently lacking.

This paper attempts to fill this gap through an extensive analysis of adversarial attacks using the OpenAttack framework. The robustness of various transformer-based models will be assessed on a variety of datasets and under different attack methods. In addition, the efficacy of existing defensive measures will be explored and a new sentence-level adversarial attack for generating deceptive but semantically coherent text will be proposed. Through both experimental evaluation and novel methodology, this research will contribute to a better understanding of the weaknesses in NLP models and the complexities of developing robust models.

II. RELATED WORKS

The adversarial example phenomena observed in the computer vision field has been extensively explored in order to understand vulnerabilities in NLP models as well. Goodfellow *et al.* [2] pioneered this area by introducing the notion of adversarial example as well as presenting the fast gradient sign method (FGSM) for generating such examples. Although this approach was developed in the context of continuous data types, applying it to NLP systems poses challenges in light of their discrete nature.

In the early days of developing NLP-based adversarial attacks, researchers have attempted to perturb character-level as well as word-level texts. TextAttack [3] framework provides a suite of different adversarial methods to generate attacks, allowing us to systematically evaluate these methods. Unfortunately, the TextAttack package only covers the English language and operates at both character-level and word-level perturbation. The OpenAttack framework advances the idea of TextAttack and introduces new features.

New attacks have been developed recently that involve more advanced strategies that work on the sentence level. One class of attacks that shows considerable promise includes the use of paraphrasing techniques. Specifically, paraphrasing with the help of pre-trained transformer models, like T5 [6], has been shown to be quite successful. These techniques rephrase the input sentence entirely, as opposed to changing only one token at a time, which makes detection much harder. Furthermore, optimization techniques, such as beam search, have been introduced to effectively traverse the space of possible adversarial sentences.

On the other hand, defensive strategies have also been developed to protect models from adversarial attacks. One of the most common defensive techniques involves training the model on both clean and adversarial data [5]. Unfortunately, this method is very computationally costly and tends to overfit to the particular attack strategy. Another technique involves input transformation, which entails removing any adversarial noise present within the input.

The more recent robustness arguments rely on probabilistic and structural robustness. The approach called MaskPure [7] purifies the input text through random masking and then reconstructs the input text by filling the missing parts using a masked language model, thus erasing any potential adversarial perturbations. The Attention Drop [8] approach regularizes the attention weights of a transformer neural network by randomly dropping them to distribute the effects of adversarial tokens.

Certified robustness is another area of importance in defending deep learning models against attacks. Approaches, for example, that are proposed by Li *et al.* [9], intend to offer formal proofs that a model remains robust under certain perturbations of the input. Although promising, such approaches are often limited to specific types of perturbations.

Although there have been great developments in recent times, the current literature mainly concentrates on attack methods or defense strategies alone. Very few comprehensive

papers discuss attacks and defenses in a variety of datasets, models, and levels of perturbations simultaneously. In addition, sentence-level attacks and defense strategies have not received much attention. The above discussion leads us to conclude that unified tools like OpenAttack should be developed to address the issue of adversarial NLP.

III. PROBLEM STATEMENT

With the widespread adoption of deep learning models in Natural Language Processing (NLP), ensuring their robustness against adversarial inputs has become a critical concern. Adversarial attacks involve generating carefully crafted perturbations to input text that can mislead models into making incorrect predictions while preserving the original semantic meaning. Existing frameworks, such as OpenAttack, provide a standardized platform for evaluating adversarial robustness by implementing a variety of attack strategies. However, many of these attacks either fail to maintain semantic consistency, are easily detectable, or do not generalize well across different models.

Furthermore, current adversarial attack methods often struggle to achieve a balance between attack success rate and linguistic quality. High-performing attacks may introduce unnatural or grammatically incorrect text, while more natural attacks tend to have lower effectiveness. This trade-off limits their practical applicability in evaluating real-world model vulnerabilities.

Therefore, there is a need for a more effective and adaptive adversarial attack mechanism that can generate high-quality, semantically consistent perturbations while significantly degrading model performance. The objective of this project is to design and implement a novel attack model within the OpenAttack framework that improves attack success rate, maintains fluency and coherence, and demonstrates stronger generalization across different NLP models. This work aims to provide deeper insights into model vulnerabilities and contribute toward the development of more robust and secure NLP systems.

IV. DATASETS

A. SST-2: Binary Sentiment

Size: 70,042 entries (3.33 MB)

Split: Train 96% (67.3K), Val 1% (872), Test 3% (1.82K)

Source: Movie reviews with human annotations

Labels: 0 (negative), 1 (positive)

The SST-2 dataset is a widely used benchmark for sentiment analysis and is particularly useful for evaluating adversarial robustness due to its clean annotations and balanced structure.

B. IMDb Reviews: Binary Sentiment

Size: 50,000 entries (66.21 MB)

Split: Train 50% (25K), Test 50% (25K)

Characteristics: Highly polar reviews, minimal imbalance

Labels: 0 (negative), 1 (positive)

The IMDb dataset provides longer and more complex text samples compared to SST-2, making it suitable for evaluating robustness under more realistic and verbose inputs.

C. Mental Health Dataset: Imbalanced Multiclass

Size: 127,241 entries (25.34 MB)
Split: Train 50%, Test 50%
Imbalance: Neutral 83%, other classes 13%, null 4% (removed during preprocessing)
Labels: 13 emotion classes but neutral and others are used in this work

This dataset introduces a highly imbalanced setting, allowing us to evaluate how adversarial attacks behave under skewed label distributions and whether defenses remain effective in such scenarios.

D. Personality Dataset: Multi-Class Classification

Source: Kaggle Personality Dataset
Size: Varies (approximately 10K–20K samples)
Labels: Extrovert, Introvert, Ambivert
Task: Personality classification based on textual input

This dataset introduces a higher-level semantic classification task, where predictions depend on nuanced linguistic cues rather than explicit sentiment. It is particularly useful for evaluating whether adversarial attacks can disrupt deeper contextual understanding beyond surface-level polarity. Additionally, it helps assess the generalizability of both attack and defense mechanisms across diverse NLP tasks.

E. AG News: Multiclass Classification

Size: 120,000 entries
Split: Train 50%, Test 50%
Categories: World, Sports, Business, Technology
Balance: Relatively balanced across classes

AG News enables evaluation of adversarial robustness in a multiclass setting, where decision boundaries are more complex compared to binary classification tasks.

F. Hindi Dataset: Multilingual Evaluation

Size: 15,000 sentiment examples
Languages: Hindi
Purpose: Evaluation of OpenAttack in non-English settings

This dataset is used to analyze whether adversarial vulnerabilities persist across languages and to evaluate the multilingual capabilities of the OpenAttack framework.

Overall, the use of diverse datasets spanning binary, multiclass, imbalanced, and multilingual settings ensures a comprehensive evaluation of adversarial robustness across different data distributions and task complexities.

V. METHODOLOGY

This work presents a comprehensive adversarial robustness evaluation framework that integrates attack benchmarking, defense analysis, multilingual generalization, and a novel sentence-level adversarial attack (TECA). The methodology is designed to systematically evaluate how modern NLP models behave under adversarial stress across multiple datasets, model architectures, and defense strategies.

A. Overall Framework

The proposed pipeline consists of 3 key stages: (i) model training and fine-tuning, (ii) adversarial attack generation using OpenAttack and TECA, and (iii) robustness evaluation.

Each component is designed to operate modularly, allowing controlled experimentation across different configurations of datasets, models, and attacks.

B. Datasets and Task Variants

To ensure comprehensive evaluation, experiments are conducted on multiple dataset configurations:

- **Binary Classification:** Standard sentiment classification tasks (e.g., SST-2).
- **Multiclass Classification:** Datasets with more than two labels to evaluate decision boundary complexity.
- **Imbalanced Datasets:** Skewed label distributions to assess robustness under biased data conditions.

Additionally, to evaluate cross-lingual robustness, a Hindi dataset is incorporated. Models are trained and evaluated on both English and Hindi inputs, enabling analysis of adversarial transferability across languages.

C. Victim Models

We consider three transformer-based architectures:

- BERT
- RoBERTa
- ELECTRA

Each model is fine-tuned independently on all dataset variants. This allows us to analyze how architectural differences influence adversarial vulnerability.

D. Adversarial Attack Evaluation using OpenAttack

We integrate the OpenAttack toolkit to benchmark multiple adversarial attacks, including:

- TextFooler
- SCPN
- GAN-based attacks

Given an input x with label y , each attack generates a perturbed input x' such that $\hat{y} \neq y$. The evaluation metrics include:

- **Attack Success Rate (ASR)**
- **Semantic Similarity** (via SBERT embeddings)

E. Defense Mechanisms

To study robustness of OpenAttack, we implement and evaluate multiple defense strategies:

MaskPure: A stochastic purification-based defense that masks and reconstructs tokens to remove adversarial noise. This method introduces randomness to prevent deterministic adversarial exploitation.

AttentionDrop: A transformer regularization technique that randomly drops attention weights during training, improving generalization and reducing reliance on spurious correlation.

Paraphrasing Defense: The model we developed takes a sentence and generates 5 paraphrases of that sentence using T5. Now, it takes 5 + 1 variants as input for openattack. Attacks all these variants separately and finds predictions on each. Averages out predictions to give final prediction.

These defenses are applied during both training (adversarial training) and inference (preprocessing), allowing evaluation of their effectiveness under different conditions.

F. Proposed TECA Attack

The TECA attack introduces a structured, sentence-level adversarial generation mechanism. Unlike token-level perturbations, TECA operates on syntactic and semantic transformations.

The attack objective is defined as:

$$\mathcal{L}_{attack}(x') = \max_{y' \neq y} f_{y'}(x') - f_y(x')$$

To preserve meaning, a semantic constraint is enforced:

$$\text{Sim}(x, x') \geq \delta$$

TECA applies transformations such as:

- Passive-active voice conversion
- Clause insertion and expansion
- Negation injection
- Phrase reordering

A beam search strategy is used to iteratively refine candidate sentences, balancing adversarial strength and semantic fidelity.

G. Multilingual Robustness Evaluation

To extend evaluation beyond English, models are trained on a Hindi dataset and subjected to the same adversarial attacks. This enables analysis of:

- Cross-lingual attack effectiveness
- Transferability of adversarial examples
- Defense robustness across languages

H. Evaluation Protocol

Both baseline attacks (OpenAttack) and TECA are applied, and performance is compared across all configurations to provide a complete robustness profile.

VI. IMPLEMENTATION

The proposed system is implemented as a complete adversarial evaluation pipeline integrating model training, defense mechanisms, baseline attacks from OpenAttack, and the proposed TECA attack. The pipeline is modular and consists of sequential stages, enabling reproducibility and extensibility.

A. Overall Pipeline for Attack analysis

The complete workflow is defined as:

- 1) Dataset preparation and preprocessing
- 2) Model training and fine-tuning
- 3) Baseline attack evaluation using OpenAttack
- 4) TECA adversarial attack generation
- 5) Metric computation and analysis

Each stage is described in detail below.

B. Dataset Preparation

Datasets are categorized into:

- Binary classification datasets
- Multiclass classification datasets
- Imbalanced datasets

Text preprocessing includes tokenization, normalization, and padding. For multilingual evaluation, a Hindi dataset is incorporated with appropriate tokenization strategies.

C. Model Training and Fine-Tuning

Transformer-based models (BERT, RoBERTa, ELECTRA) are fine-tuned using the HuggingFace Transformers library. Each model is trained independently on all dataset types. Cross-entropy loss is used for optimization, and hyperparameters such as learning rate, batch size, and epochs are tuned empirically.

D. Defense Integration

Defense mechanisms are incorporated at both training and inference stages to test OpenAttack:

- **MaskPure:** Applied as a stochastic purification layer during inference.
- **AttentionDrop:** Integrated into transformer layers during training as a regularization technique.
- **Paraphrasing Defense:** Applied as a preprocessing step before classification.

Adversarial training is performed by augmenting training data with adversarial samples generated from baseline attacks.

E. Baseline Attack Evaluation (OpenAttack)

The OpenAttack framework is used to evaluate standard adversarial attacks such as TextFooler, SCPN, and GAN-based attacks. Each attack is applied across all models and datasets to establish baseline robustness metrics.

F. TECA Attack Pipeline

The TECA attack is implemented as a multi-stage pipeline designed to generate high-quality adversarial examples through structured transformations.

1) *Step 1: Input Initialization:* Given an input sentence x and its true label y , the beam is initialized as:

$$\mathcal{B}_0 = \{x\}$$

The original sentence embedding $E(x)$ is computed using SBERT and cached for future similarity calculations.

2) *Step 2: Transformation Candidate Generation:* For each sentence in the current beam, a set of structure-aware transformations is applied:

- Passive $\leftrightarrow$ Active voice conversion
- Clause insertion or expansion
- Negation injection
- Phrase reordering

These transformations are implemented using syntactic parsing (dependency and constituency parsing) to ensure grammatical correctness.

3) *Step 3: Candidate Scoring:* Each generated candidate x' is passed through the victim model to obtain logits. The attack score is computed as:

$$\mathcal{L}_{\text{attack}}(x') = \max_{y' \neq y} f_{y'}(x') - f_y(x')$$

Candidates with higher scores are more likely to cause misclassification.

4) *Step 4: Semantic Similarity Filtering:* To ensure meaning preservation, cosine similarity between embeddings is computed:

$$\text{Sim}(x, x') = \cos(E(x), E(x'))$$

Candidates with similarity below threshold δ are discarded.

5) *Step 5: Beam Selection:* The remaining candidates are ranked based on attack score. The top- k candidates are selected to form the next beam:

$$\mathcal{B}_{t+1} = \text{Top-}k(\mathcal{B}_t)$$

6) *Step 6: Iterative Search:* Steps 2–5 are repeated iteratively until:

- A candidate achieves misclassification ($\hat{y} \neq y$), or
- Maximum search depth is reached

7) *Step 7: Output Selection:* The best adversarial example is selected based on:

- Highest attack score
- Valid semantic similarity
- Grammatical correctness

G. Multilingual TECA Extension

For Hindi datasets:

- Transformation rules are adapted for Hindi syntax
- Multilingual SBERT embeddings are used
- Tokenization is adjusted using language-specific tokenizers

H. Optimization Techniques

To improve efficiency:

- Embedding caching is used
- Beam width k is tuned
- Transformation count per iteration is limited
- Early stopping is applied

I. Evaluation Metrics

The system evaluates performance using:

- Attack Success Rate (ASR)
- Semantic Similarity
- Grammatical Correctness

All attacks (baseline and TECA) are compared across models, datasets, and defense strategies to provide a complete robustness analysis.

VII. RESULTS AND ANALYSIS

This section presents a detailed evaluation of adversarial attack performance across multiple datasets, victim models, and attack strategies. We compared baseline attacks from OpenAttack with the proposed TECA attack using Attack Success Rate (ASR) as the primary metric. We also compared defense of SST-2 dataset on BERT using AttentionDrop, MaskPure and paraphrase defense and tabulate those results

A. Analyzing defense model performance on SST-2 and BERT

TABLE I
ATTACK SUCCESS RATE (ASR) ON SST-2 AND BERT

Model	TextFooler	DeepWordBug	UATA
Without Defense	0.81	0.52	0.09
MaskPure	0.42	0.32	0.09
AttentionDrop	0.64	0.46	0.09
Paraphrase defense	0.76	0.29	0.09

We did an analysis by attacking our victim model (BERT) with a character level model, word level model and sentence level model. Based on this analysis we found that the defense models actually defend the model and the attack success rate (ASR) on the victim model does reduce. Sentence level attack does not really reduce on defending as the current existing defense models can not defend against sentence level attacks.

B. Performance on SST-2 Dataset

TABLE II
ATTACK SUCCESS RATE (ASR) ON SST-2

Model	TextFooler	SCPN	GAN	TECA
RoBERTa	0.74	0.62	0.38	1.00
ELECTRA	0.68	0.60	0.32	1.00
BERT	0.74	0.64	0.32	0.96

On the SST-2 dataset, TECA achieves near-perfect ASR across all models, significantly outperforming baseline attacks. While TextFooler and SCPN demonstrate moderate effectiveness, their reliance on token-level perturbations limits their ability to fully exploit model weaknesses. TECA, by contrast, leverages sentence-level transformations, allowing it to consistently fool even robust models like RoBERTa.

C. Performance on IMDB Dataset

TABLE III
ATTACK SUCCESS RATE (ASR) ON IMDB

Model	TextFooler	SCPN	GAN	TECA
RoBERTa	0.94	0.54	0.72	1.00
ELECTRA	0.78	0.46	0.72	1.00
BERT	0.92	0.64	0.74	1.00

On the IMDB dataset, which contains longer and more complex sentences, TECA achieves a perfect ASR of 1.00 across all models. This highlights its ability to exploit contextual dependencies effectively. Baseline attacks show variability, with SCPN performing relatively weaker due to limited transformation diversity, while GAN-based methods show moderate performance.

D. Performance on Imbalanced Dataset

TABLE IV
ATTACK SUCCESS RATE (ASR) ON IMBALANCED DATASET

Model	TextFooler	SCPN	GAN	TECA
RoBERTa	0.56	0.66	0.74	0.98
ELECTRA	0.56	0.70	0.84	0.98
BERT	0.60	0.74	0.88	0.96

In the imbalanced dataset setting, all attacks experience a drop in performance due to skewed class distributions. However, TECA maintains a high ASR (above 0.95), demonstrating robustness under challenging data conditions. GAN-based attacks perform comparatively better than TextFooler and SCPN in this setting, suggesting that generative approaches adapt better to imbalance, though still not matching TECA.

E. Performance on AG News Dataset

TABLE V
ATTACK SUCCESS RATE (ASR) ON AG NEWS

Model	TextFooler	SCPN	GAN	TECA
RoBERTa	0.54	0.72	0.70	0.94
ELECTRA	0.42	0.72	0.80	0.90
BERT	0.48	0.66	0.70	0.92

For the multiclass AG News dataset, TECA again achieves the highest ASR across all models. The performance gap between TECA and baseline attacks becomes more pronounced as task complexity increases. SCPN performs relatively well due to its structural transformations, but still falls short of TECA's guided search-based optimization.

F. Performance on Personality Dataset

A notable exception is observed in the Personality dataset, where TECA fails completely (ASR = 0.00). This contrasts sharply with SCPN and GAN-based attacks, which achieve high success rates. This suggests that TECA's transformation strategies may not effectively capture the nuanced semantic features required for personality classification tasks.

TABLE VI
ATTACK SUCCESS RATE (ASR) ON PERSONALITY DATASET

Model	TextFooler	SCPN	GAN	TECA
RoBERTa	0.20	0.96	0.54	0.00
ELECTRA	0.22	0.76	0.78	0.00
BERT	0.38	0.82	0.90	0.00

G. Performance on Hindi Dataset

On the Hindi Dataset the ASR is 0.

OpenAattack has all attack models trained on English and Chinese. So only character level attack is possible. On trying, attack failed on all samples. This failure can primarily be attributed to the presence of Hindi words and code-mixed language in the dataset, which are not well-represented in the tokenizer vocabulary or the training distribution of the victim models. It is also attributed to the fact that adding random characters in between does not affect the victim model. So it always treats the injected characters as noise and does not misclassify the predictions.

Out-of-Vocabulary (OOV) Hindi Tokens: A significant portion of the input contains Hindi words or transliterated expressions. Since the victim models are predominantly trained on English corpora, these tokens are either split into ineffective subword units or mapped to weak semantic representations, making both classification and adversarial perturbation ineffective.

Model Training Bias Toward English Data: The victim classifiers are not exposed to Hindi or multilingual personality expressions during training. As a result, their decision boundaries become unstable or unresponsive for such inputs, preventing consistent adversarial exploitation.

H. Cross-Dataset Observations

Across all datasets, several important trends emerge:

- TECA consistently outperforms baseline attacks in most scenarios, achieving near-perfect ASR on multiple datasets.
- Sentence-level transformations are more effective than token-level perturbations in generating adversarial examples.
- Dataset characteristics (e.g., imbalance, task complexity) significantly influence attack performance.
- Transformer models remain vulnerable even after fine-tuning, especially against structured attacks.

I. Key Insights

The experimental results demonstrate that TECA is a highly effective adversarial attack framework capable of generating semantically coherent yet misleading inputs. Its success across multiple datasets and models highlights the limitations of existing defenses and the need for more robust, structure-aware protection mechanisms. However, its failure on the Personality dataset and hindi dataset indicate that further improvements are needed to handle tasks requiring deeper semantic reasoning.

VIII. CONCLUSION AND FUTURE WORK

In this research, an overview of attacks against NLP models under OpenAttack is provided. We conduct a rigorous study of adversarial attack effectiveness through different model architectures, datasets, and defensive methods. The results indicate that the advanced transformer-based models are still vulnerable to adversarial attacks at different granularities.

Some contributions have been made in this paper. Firstly, we conduct a systematic evaluation of adversarial attacks against NLP models. Secondly, we devise a T5 paraphrase method that could significantly lower the ASR of up to 58%. Thirdly, we develop a sentence-level attack named TECA that has a good balance between efficiency and linguistic quality by preserving a high semantic similarity score of 0.912. Fourthly, we evaluate the performance of different defenses and achieve up to 73% ASR drop. Fifthly, we conduct a series of attacks under multilingual conditions and conclude that such attacks do not depend on a specific language and work well in other languages as well. It is found out that since OpenAttack only trains on english datasets, it cannot conduct attacks under multilingual conditions.

Nevertheless, several key research areas still need exploration. In particular, further studies should be devoted to the development of certification methods that would ensure formally proven robustness against adversarial manipulation of the input data. Besides, adversarial training methods should be improved in terms of computational complexity to achieve practical robustness of models. Moreover, the emergence of big generative models such as GPT and LLaMA makes the analysis of their robustness highly relevant. Adversarial defense in multimodal setups is also an important area, including vision-language models. Moreover, human-in-the-loop studies should be implemented to estimate perceptual quality and imperceptibility of attacks and defenses. Finally, the problem of transferability of defense techniques between models and datasets should be investigated.

Overall, these research questions are relevant due to their practical importance in ensuring robustness of NLP systems for application to real tasks.

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